Mixed-Unit Refresher: Parent Key

November 4, 2026 | Worked solutions and scoring | Keep separate from quiz

1. **20 points**

(a) Factor and simplify:

$$10x^2 - 3x - 4 = (5x - 4)(2x + 1)$$

$$6x^2 + 15x + 6 = 3(2x + 1)(x + 2)$$

$$R(x) = \frac{5x - 4}{3(x + 2)}, \qquad x \neq -2, -\frac{1}{2}$$

Domain: $(-\infty, -2) \cup (-2, -\frac{1}{2}) \cup (-\frac{1}{2}, \infty)$. Canceling does not restore an excluded input.

(b) x-intercept: $(\frac{4}{5}, 0)$. The canceled zero is not an intercept.
y-intercept: $R(0) = -\frac{4}{6} = -\frac{2}{3}$, so $(0, -\frac{2}{3})$.
Vertical asymptote: $x = -2$. Horizontal asymptote: $y = \frac{10}{6} = \frac{5}{3}$.

For the hole, substitute the canceled zero into the simplified expression:

$$\frac{5(-\frac{1}{2}) - 4}{3(-\frac{1}{2} + 2)} = -\frac{13}{9} \quad \Longrightarrow \quad \left(-\frac{1}{2}, -\frac{13}{9}\right)$$

(c) $R(x) = \frac{5}{3} - \frac{14}{3(x+2)}$. Draw dashed lines at $x = -2$ and $y = \frac{5}{3}$, the open circle, and both intercepts. Left of -2, the branch is above $y = \frac{5}{3}$ and tends to $+\infty$ near -2. Right of -2, it is below $y = \frac{5}{3}$ and tends to $-\infty$ near -2.

Scoring (20): domain and simplification 4; intercepts 4; hole 3; asymptotes 3; sketch 4; one-sided explanation 2.

Use of this key: A suggested 90/100 mastery target is a local diagnostic benchmark, not Kai's school rubric. An Opus 5.5 simulated Honors Precalculus A-student attempt reviewed the draft against the November 1 scans; it is not a prediction of Kai's grade.

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2. **20 points**

(a) Divide leading terms, multiply, and subtract:

$$\frac{6x^3}{3x^2} = 2x$$

$$2x(3x^2 + x - 2) = 6x^3 + 2x^2 - 4x$$

$$(6x^3 - 7x^2 - 2x + 2) - (6x^3 + 2x^2 - 4x) = -9x^2 + 2x + 2$$

The next term is:

$$\frac{-9x^2}{3x^2} = -3$$

$$-3(3x^2 + x - 2) = -9x^2 - 3x + 6$$

$$(-9x^2 + 2x + 2) - (-9x^2 - 3x + 6) = 5x - 4$$

$$S(x) = 2x - 3 + \frac{5x - 4}{3x^2 + x - 2}$$

The slant asymptote is $\boxed{y = 2x - 3}$, not the remainder.

(b) As $x \to -\infty$, $S(x) \to -\infty$. As $x \to +\infty$, $S(x) \to +\infty$. The difference $S(x) - (2x - 3)$ tends to zero because its numerator degree is smaller than its denominator degree. Thus the ends approach that line. For sufficiently negative inputs the graph is below the line; for sufficiently positive inputs it is above the line.

Scoring (20): long division 10; asymptote 4; end directions 2; explanation that the remainder fraction tends to zero 4. Above/below detail is useful but not required for full credit.

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3. **20 points**

(a) Domain: $(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$. Exclude the vertical asymptote and the hole.

(b) One possible equation is:

$$K(x) = \frac{2(x-3)(x-1)}{(x+1)(x-1)}$$

The factor $x - 3$ gives the x-intercept 3; $x + 1$ gives vertical asymptote $x = -1$; equal degrees and leading ratio 2 give $y = 2$. The canceled $x - 1$ factor preserves a hole rather than an intercept.

To see the scale, start with $A(x-3)/(x+1)$. Its horizontal asymptote is $y = A$, so $A = 2$. The original function must still be undefined at $x = 1$.

$$\text{Hole height: } \frac{2(1-3)}{1+1} = -2$$

$$\text{y-intercept: } \frac{2(0-3)}{0+1} = -6$$

Accept algebraically equivalent expressions with the same graph and domain. In particular, $2 - \frac{8}{x+1}$ is acceptable only if the student explicitly excludes $x = 1$ as well as $x = -1$. No unique written form is required.

(c) $K(x) = 2 - \frac{8}{x+1}$. As $x \to -\infty$, the graph approaches 2 from above. As $x \to +\infty$, it approaches 2 from below.

Scoring (20): domain 4; valid equation including excluded point 8; explanation of features 4; end behavior 4. Do not penalize a different equivalent equation.

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4. **20 points**

(a) $T(8) = 23$: an 8-page job takes 23 seconds.
$T^{-1}(8) = 2$: an 8-second job contains 2 pages.
Both rows contain 8, but the lookups have different directions.

(b) Find the rate and setup time:

$$\frac{13-8}{4-2} = \frac{5}{2} \text{ seconds per page}$$

$$8 = \frac{5}{2}(2) + b \quad \Longrightarrow \quad b = 3$$

$$T(p) = \frac{5}{2}p + 3$$

To invert, solve for pages:

$$t = \frac{5}{2}p + 3 \quad \Longrightarrow \quad t - 3 = \frac{5}{2}p$$

$$T^{-1}(t) = \frac{2}{5}(t-3)$$

The inverse takes total seconds as input and returns pages.

$$\text{(c)} \quad T^{-1}(53) = \frac{2}{5}(53-3) = 20$$

A 53-second job contains **20 pages**. Check: $\frac{5}{2}(20) + 3 = 53$ seconds.

Scoring (20): two correct lookups and meanings 6; forward model 4; inverse model and algebra 4; new inverse evaluation, work, and units 6. Do not give full interpretation credit for a number without units.

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5. **20 points**

(a) Solve for the original input:

$$y = 2 + \sqrt{5x + 10} \quad \Longrightarrow \quad y - 2 = \sqrt{5x + 10}$$

$$(y - 2)^2 = 5x + 10 \quad \Longrightarrow \quad x = \frac{(y - 2)^2 - 10}{5}$$

$$U^{-1}(x) = \frac{(x - 2)^2 - 10}{5}$$

Domain of U: $[-2, \infty)$, since $5x + 10 \geq 0$. Range of U: $[2, \infty)$, so the inverse domain is $[2, \infty)$.

(b) Both expected results are x, on the appropriate domains.

$$\begin{aligned} U(U^{-1}(x)) &= 2 + \sqrt{5\left(\frac{(x - 2)^2 - 10}{5}\right) + 10} \\ &= 2 + \sqrt{(x - 2)^2} \\ &= 2 + |x - 2| = x, \qquad x \geq 2. \end{aligned}$$

$$\begin{aligned} U^{-1}(U(x)) &= \frac{\left[(2 + \sqrt{5x + 10}) - 2\right]^2 - 10}{5} \\ &= \frac{(5x + 10) - 10}{5} = x, \qquad x \geq -2. \end{aligned}$$

The first restriction matters: $\sqrt{z^2} = |z|$. Also, the first composition cannot take $x < 2$ because that input is outside the inverse function's domain.

Scoring (20): inverse formula and algebra 6; both domains 4; expected composition results 2; algebra for each order 3 + 3; allowed inputs for each order 1 + 1.